TRANSFORMING CRITICAL CARE: MACHINE LEARNING MODELS FOR REAL-TIME ACUTE KIDNEY INJURY PREDICTION

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INTRODUCTION: Acute Kidney Injury (AKI) affects up to 50% of ICU patients, significantly increasing mortality and healthcare costs. Current gold standards (serum creatinine and urine output) are lagging indicators as by the time these values change, kidney damage has already occurred. Our objective was to build a machine learning model that predicts AKI in advance, helping shift clinical management from reactive treatment toward proactive prevention.

METHODS: We used the MIMIC-IV database, starting with approximately 75,000 ICU stays cleaned into a high-fidelity adult cohort. Cleaning involved removing impossible outliers (urine output > 50 L/day, weights < 30 kg, negative heart rates) and applying physiological boundaries. The final feature set included 46 variables across seven domains: demographics, labs, vitals, fluid balance, hemodynamics, neurologic status, and comorbidities. We trained XGBoost, LSTM Network, Random Forest, and Logistic Regression and combined their predictions through ensemble learning. The 75/25 class imbalance was addressed using cost-sensitive weighting (scale_pos_weight), and the decision threshold was tuned to 0.45 to maximise Recall, preferring a false alarm over a missed diagnosis.

RESULTS: The weighted ensemble model achieved an AUROC of 0.8884, AUPRC of 0.7707, Recall of 0.8123, Precision of 0.5738, and an F1-score of 0.6725. SHAP analysis identified urine output (ml/kg-hr), delta-creatinine, mean arterial pressure, heart rate, and BUN as the most important predictors. Rising delta-chloride values over a 24-hour period showed a direct correlation with AKI onset. Multi-organ indicators like increasing SOFA score, decreasing GCS, lower MAP, and elevated bilirubin further reinforced the clinical relevance of the predictions.

CONCLUSIONS: Our ensemble machine learning model offers a promising tool for early AKI prediction in critically ill patients. Embedded within Electronic Health Record systems, it could enable real-time monitoring of labs, vitals, and urine output, serving as decision support that complements rather than replaces clinical judgement.